METRIC
MIL-PRF-8565/13A
w/AMENDMENT 1
13 December 2013
SUPERSEDING
MIL-PRF-8565/13A
14 July 2009

PERFORMANCE SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, MEDIUM-RATE, TYPE 1, MAINTENANCE-FREE, 24-VOLT, 10-AMPERE-HOUR

Reactivated after 14 July 2009 and may be used for new and existing designs and acquisitions.

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-8565.

REQUIREMENTS:

1. Part or identifying number. The part or identifying number (PIN) of the batteries covered by this specification sheet shall be D8565/13-1.

2. Dimensions and weight. The dimensions of the battery shall be as shown on figure 1. The weight of the battery shall be not greater than 14.03 kilograms (31.0 pounds) (see 3.8.3 and 4.5.3 of MIL-PRF-8565).

3. Capacity. The rated capacity of the battery shall be not less than 10 ampere-hours (Ah), based on its ability to conform to the life test final discharge. The life test final discharge is a 1-hour discharge at 24°C ambient temperature with a 10-ampere load (1 C-rate) to a cutoff voltage of 18.0 volts (10.0 Ah/1 Hr/24°C/18.0 V). See 3.8.6, 4.5.7, and 6.10 of MIL-PRF-8565.

4. Service life. The battery's service life (useful life) in aircraft shall be not less than 4 years.

5. Qualification inspection and conformance inspection. Table I lists the required inspections.

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6. MIL-PRF-8565 variance. The battery shall comply with MIL-PRF-8565 except as follows.

6.1 The following paragraphs are not applicable:

3.8.5 Vent tubes.
3.8.7 Constant voltage discharge.
3.8.8 Strength of receptacle.
3.8.13.2 Shelf life.
3.8.13.3 Deep discharge recovery.
3.8.14 Impact resistance (nonmetallic battery container only).
3.8.15 Shock (crash safety).
3.8.17 Cold and hot temperature charge.
4.5.5 Strength of vent tubes.
4.5.8 Constant voltage discharge.
4.5.9 Strength of receptacle.
4.5.24 Impact resistance test (nonmetallic container only).
4.5.25 Shock test (crash safety).
4.5.26 Shelf life test.
4.5.27 Deep discharge recovery test.
4.5.29 Cold and hot temperature charge test.
4.5.30 Wormgear retaining pin hardness.

6.2 Modify the following paragraphs:

2.3 Non-government publications. Add the following:

"ARINC Specification 404	-	Air Transport Equipment Cases and Racking

(Copies of this specification are available from www.arinc.com or ARINC Incorporated, 2551 Riva Road, Annapolis, MD 21401.)"

3.5.3 Battery container and cover. Add the following after the last sentence in the paragraph: "The battery container shall meet the requirements of ARINC Specification 404 for a ½ ATR short case size, except as modified herein".

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3.5.4 Venting. Delete MIL-PRF-8565 paragraph in its entirety and substitute the following: "The battery case shall contain one or more holes for venting of gases, as depicted on figure 1 of MIL-PRF-8565/13. Vents shall be located such that the container cover, battery monoblocs, or cell assemblies do not block the venting holes. If the hole(s) size exceeds 6.5 millimeters, the hole(s) shall be covered by a plastic or wire screen to prevent debris from entering the battery. The cells shall be equipped with a resealable venting mechanism to prevent excessive internal pressure build-up. When the battery is operating within the requirements of this specification, cell venting shall not cause the battery to discharge entrained electrolyte."

3.5.5 Receptacles. Add the following: "The receptacle for the battery's positive and negative connections shall be a Cannon part number DPXB-8-34P-0101-A152 or equivalent. The receptacle shall be located as shown on figure 1 of MIL-PRF-8565/13 and wiring connections shall conform to figure 2 of MIL-PRF-8565/13."

3.6.2 Polarity marking. Delete MIL-PRF-8565 paragraph in its entirety and substitute the following: "The container shall be conspicuously and durably marked with "+" and "-" in white as shown on figure 1. The receptacle reference designation "P1" shall be located as shown on figure 1, with white marking not less than 3 millimeters high. In addition, the schematic diagram of figure 2 shall be placed on label no. 3 and located as shown on figure 1. Both figure 1 and 2 are in MIL-PRF-8565/13."

3.6.4 Battery caution marking. Delete MIL-PRF-8565 paragraph in its entirety and substitute the following: "Marking for label 2 shown on figure 1 of MIL-PRF-8565/13 shall show the following. The lettering height shall be not less than 6.35 millimeters (0.25 inches). The marking shall be in black or white print.

MAINTENANCE-FREE BATTERY

DO NOT REMOVE COVER

PROCESS THIS BATTERY IN ACCORDANCE WITH APPROVED PROCEDURES ONLY

**SEE NAVAIR 17-15BAD-1
OR APPLICABLE SERVICE T.O."**

3.8.10 Functional discharge. Delete MIL-PRF-8565 paragraph in its entirety and substitute the following: "After the environmental tests of 4.5.14, 4.5.15, 4.5.16, 4.5.18, 4.5.19, 4.5.20, 4.5.21, and 4.5.23, all of MIL-PRF-8565, the battery shall be subjected to a 400-watt discharge for 5 minutes. The battery voltage shall be not less than 21.0 volts at the end of the 5 minutes."

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3.8.11 Temperature rise and float. Delete MIL-PRF-8565 paragraph in its entirety and substitute the following: "When tested in accordance with 4.5.17, as modified by MIL-PRF-8565/13, the battery voltage shall be measured at intervals of not greater than 5 minutes. The battery voltage, during the 8 hours of constant current charge at 2.0 amperes, shall be not less than 29.0 volts once 110 percent of the capacity removed during discharge has been returned. The battery shall meet requirement (1) in table III of MIL-PRF-8565/13 following the charge period."

4.4.2 Constant-potential charging method. Delete MIL-PRF-8565 paragraph in its entirety and substitute the following: "The battery shall be charged at a constant current rate of 2.00 ±0.05 amperes until the battery voltage reaches 29.0 ±0.1 volts. The charging shall then be continued for 2 hours at 29.0 ±0.1 volts."

4.5.10 Cycling test. Second sentence delete "100 cycles" and substitute "85 cycles". Third sentence delete "2.0-hour constant potential charge in accordance with 4.4.2" and substitute "charge in accordance with 4.4.2 of as modified by MIL-PRF-8565/13".

4.5.12 Charge and discharge test at low temperature. Delete steps (a) through (g) and substitute the following:

"a. Charge the battery in accordance with 4.4.2 as modified by MIL-PRF-8565/13".

"b. Place the battery in a test chamber in accordance with 4.4.1 of MIL-PRF-8565 at -18°C, then discharge at 400 watts for not less than 15 minutes.

"c. Remove the battery from the chamber and charge for 2.0 hours at 27.0 ±0.1 volts with the current source limited to 2.0 ±0.1 amperes.

"d. Immediately following charge, discharge the battery at 400 watts to 21.0 volts. Examine the battery for the requirement (3) in table III of MIL-PRF-8565/13."

4.5.13 Discharge while inverted. Delete steps c and step d and substitute the following:

"c. Discharge the battery at 400 watts for 5 minutes or to a cutoff voltage of 21.0 volts, whichever occurs first. At 2.5 minutes into the discharge, place the battery upright.

"d. Examine the battery for the requirements of 3.5.9 of MIL-PRF-8565 and requirement (6) of table III of MIL-PRF-8565/13."

4.5.17 Temperature rise and float test. Delete step c, d, e, g, and h and substitute the following:

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"c. Discharge the battery at 400 watts for 30 ±0.1 minutes.

"d. Immediately after the 400 watt discharge with the battery still in the temperature chamber, constant current charge the battery at 2.0 ±0.1 amperes for 8.0 ±0.1 hours.

"e. Stabilize the battery at 24 ± 5°C. Discharge the battery in accordance with 4.5.7.

"g. Examine the battery for the requirements of 3.7 of MIL-PRF-8565 and 3.8.11, as modified by MIL-PRF-8565/13 ."

4.5.28 Output performance test. Delete the text and substitute: "This test consists of the following steps.

"a. Charge each battery in accordance with paragraph 4.4.2, as modified by MIL-PRF-8565/13.

"b. Stabilize the three samples in accordance with 4.4.1 of MIL-PRF-8565 at the applicable test temperatures as specified in requirements (3), (4), and (5) in table III of MIL-PRF-8565/13.

"c. Discharge sample 1 in accordance with requirement (3) in table III of MIL-PRF-8565/13.
Discharge sample 2 in accordance with requirement (4) in table III of MIL-PRF-8565/13.
Discharge sample 3 in accordance with requirement (5) in table III of MIL-PRF-8565/13.

"d. Each sample shall meet the minimum cutoff voltage of the applicable requirements in table III of MIL-PRF-8565/13."

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6.3 Add the following paragraphs:

"3.5.11 Latching handles. The battery shall be equipped with two lever latch handles, Camloc part number 61L2-1-2AA or equivalent. The handles shall be located as shown on figure 1 of MIL-PRF-8565/13."

"3.5.12 Circuit breaker. The battery shall be equipped with a 20-ampere thermal overload (trip-free) circuit breaker, type MS3320-20 per SAE AS33201. The circuit breaker shall be located as shown on figure 1, and wiring connections shall conform to figure 2. Both figures 1 and 2 are in MIL-PRF-8565/13. The circuit breaker shall be mounted in a suitable housing extending from the face of the battery case, provided that the total extension (housing plus circuit breaker) does not protrude above the top surfaces of the battery when the circuit breaker is tripped and provided that the housing does not interfere with the operation of the latches."

"3.8.18 Float life. When tested in accordance with 4.5.32 of MIL-PRF-8565/13, batteries shall provide not less than rated capacity after 50 days of float charging at a temperature of 50°C and a charging rate of 40 milliamperes."

"4.5.32 Float life. The battery shall be charged at 40 milliamperes at 50 ±2°C for a period of 50 days. At the end of this period, the battery shall be stabilized at room temperature and discharged in accordance with 4.5.7 of MIL-PRF-8565. Examine the battery for the requirements of 3.8.18 of MIL-PRF-8565/13."

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TABLE I. Qualification inspection and conformance inspection of batteries.

Test number	Examination and tests	Qualification inspection						Conformance inspection		
		Sample number				Requirement paragraph 1/	Test paragraph 1/	Group A	Group B sample number	
		1	2	3	4				1	2
1	INCOMING INSPECTION	X	X	X	X	3.8.1(G)	4.5.1(G)	X	X	X
2	VISUAL AND MECHANICAL EXAMINATION	X	X	X	X	3.5(G), 3.6(G),3.8.2(G), Table II (G)	4.5.2(G)	X	X	X
3	DIMENSIONS AND WEIGHT	X	X	X		Reqmt 2(N) 3.8.3(G)	4.5.3(G)	X 2/	X	X
4	DIELECTRIC STRENGTH AND INSULATION RESISTANCE TEST	X	X	X	X	3.7(G) 3.8.4(G)	4.5.4(G)	X 2/	X	X
5	ELECTRICAL TEST PREPARATION	X	X	X	X	---	4.5.6(G)		X	X
6	CAPACITY DISCHARGE	X	X	X	X	Reqmt 3(N), 3.8.6(G)	4.5.7(G)	X	X	X
7	OUTPUT PERFORMANCE TEST	X	X	X		3.8.16(G)	4.5.28(G)		X	X
8	CHARGE AND DISCHARGE TEST AT LOW TEMPERATURE			X		Reqmt (3) table III(S)	4.5.12(S)	X		
9	CYCLING TEST	X				Reqmt (1) table III(S), 3.8.9(G)	4.5.10(S)		X	
10	FLOAT LIFE		X			3.8.18(N)	4.5.32(N)			
11	CAPACITY DISCHARGE AT 49°C (120°F)		X			Reqmt (2) table III(S), 3.8.6(G)	4.5.11(G)			X
12	DISCHARGE WHILE INVERTED			X		3.5.7(G), 3.5.9(G), Reqmt (6) table III(S)	4.5.13(S)		X	
13	ALTITUDE TEST			X		3.5.9(G), 3.7(G), 3.8.10(S)	4.5.14(G)			X
14	SHOCK (BASIC)	X				3.7(G), 3.8.10(S)	4.5.15(G)			
15	TEMPERATURE SHOCK TEST	X				3.7(G), 3.8.10(S)	4.5.16 (G)		X	
16	TEMPERATURE RISE AND FLOAT			X		Reqmt (1) table III(S), 3.7(G), 3.8.11(S)	4.5.17(S)			X
17	BATTERY GAS EMISSION TEST			X		3.8.10(S), 3.8.12(G)	4.5.18(G)			
18	VIBRATION				X	3.7(G), 3.8.10(S)	4.5.19(G)			
19	HUMIDITY TEST		X			3.7(G), 3.8.10(S)	4.5.20(G)			X
20	SALT FOG TEST	X				3.7(G), 3.8.10(S)	4.5.21(G)		X	
21	GROUND STORAGE TEST				X	3.8.13.1(G)	4.5.22(G)			
22	PHYSICAL INTEGRITY AT HIGH TEMPERATURE TEST		X			Reqmt (1) Table III(S), 3.7(G), 3.8.10(S)	4.5.23(G)			
23	FINAL EXAMINATION	X	X	X	X	3.7(G), 3.8.4(G)	4.5.31(G)		X	X

1/ The notations in parentheses specify the source of the paragraph. "(G)" means the paragraph as shown in MIL-PRF-8565. "(S)" shows that the paragraph has been modified by this specification sheet. "(N)" indicates that a new paragraph or requirement has been added by MIL-PRF-8565/13.
2/ See sample information in test paragraph.

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TABLE III. Capacity performance requirements. Delete the table and insert:

"TABLE III. Capacity performance requirements.

Requirement no.	Discharge cutoff voltage and temperature	Rate of discharge factor	Minimum time to cutoff voltage
(1)	18.0 V at 24 ± 3° C	1.0C	60 minutes
(2)	18.0 V at 49 ± 2° C	1.0C	66 minutes
(3)	21.0 V at -18 ± 2° C	400 watts 1/	20 minutes
(4)	21.0 V at 5 ± 2° C	400 watts 1/	30 minutes
(5)	21.0 V at -40 ± 2° C	400 watts	4 minutes
(6)	21.0 V at 24 ± 3°C	400 watts	5 minutes

1/ During the first minute of discharge, the discharge rate shall be 425 watts. For the remaining minutes of the discharge, the discharge rate shall be 400 watts. The minimum voltage of 21.0 volts shall apply during the entire discharge period."

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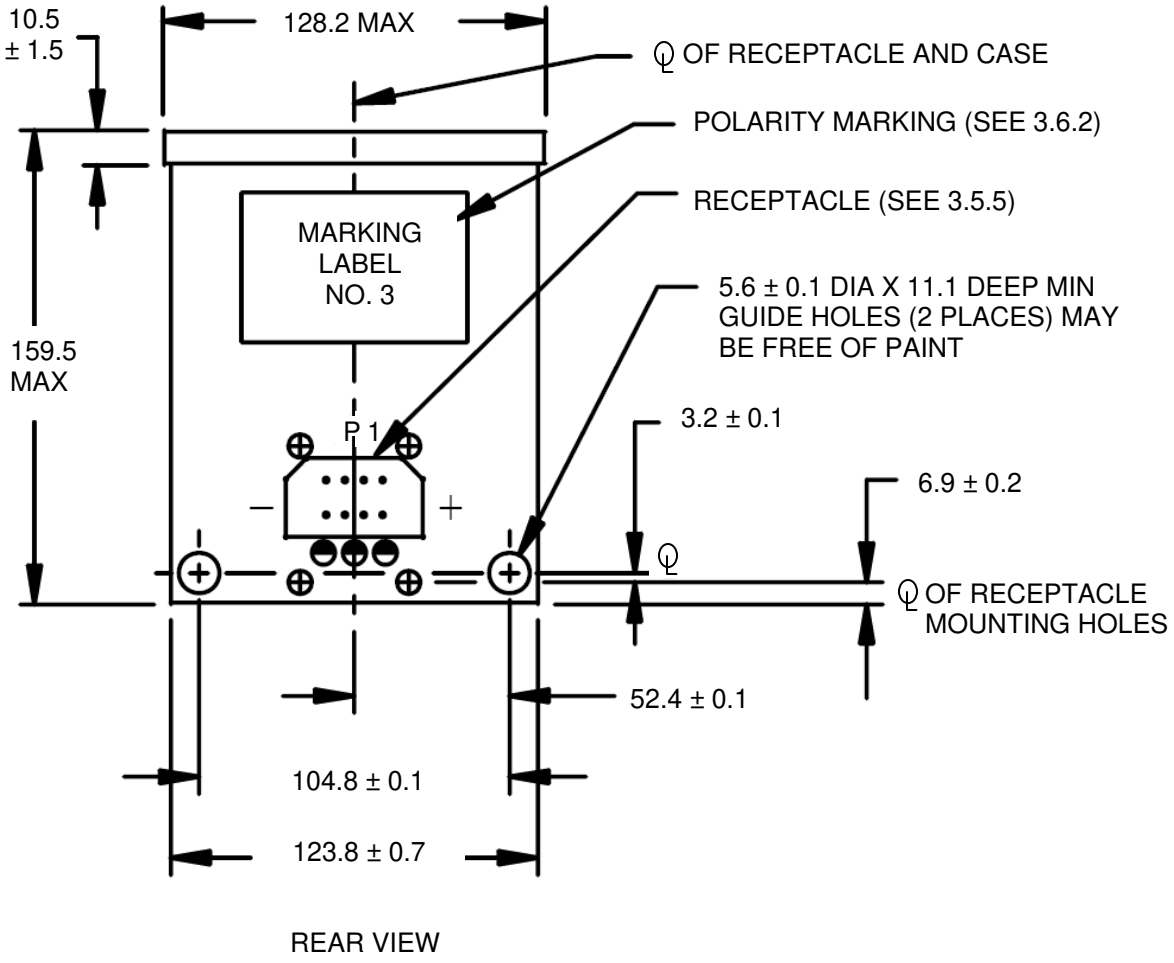


FIGURE 1. PIN D8565/13-1 dimensions.

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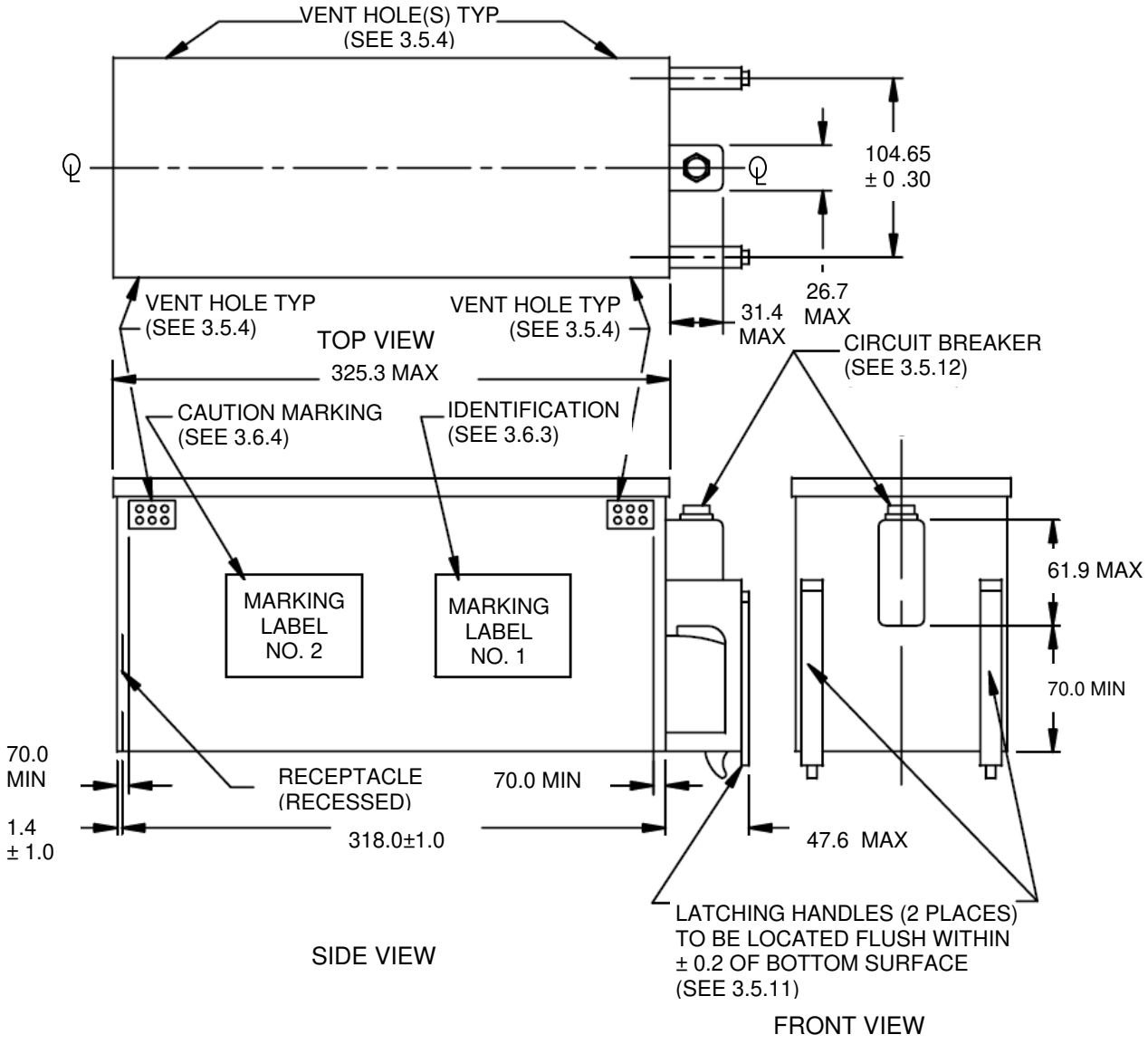


FIGURE 1. PIN D8565/13-1 dimensions - Continued.

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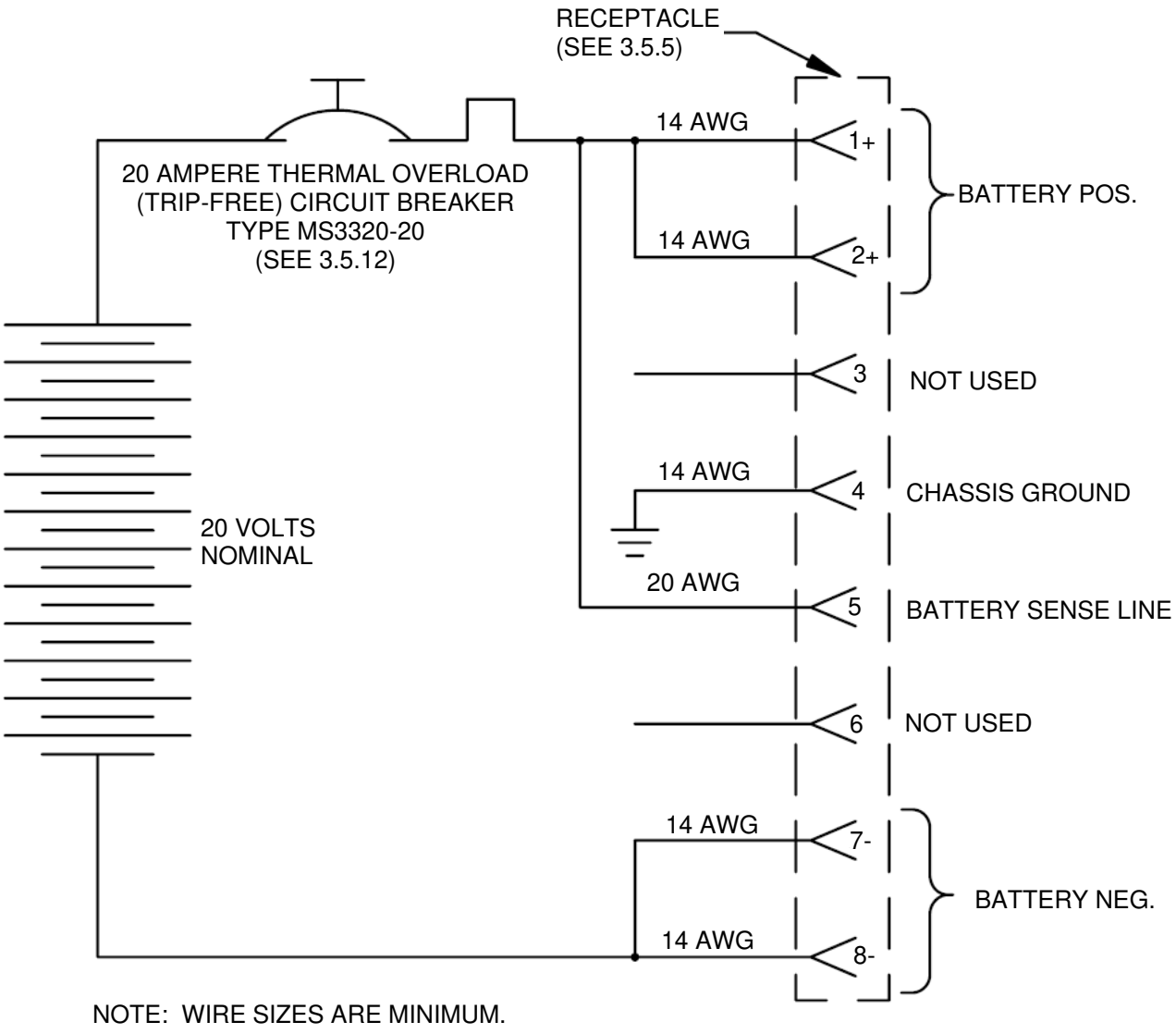


FIGURE 2. Wiring diagram.

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Referenced documents. In addition to MIL-PRF-8565, this specification sheet references SAE AS33201.

Amendment notations. The margins of this specification sheet are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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